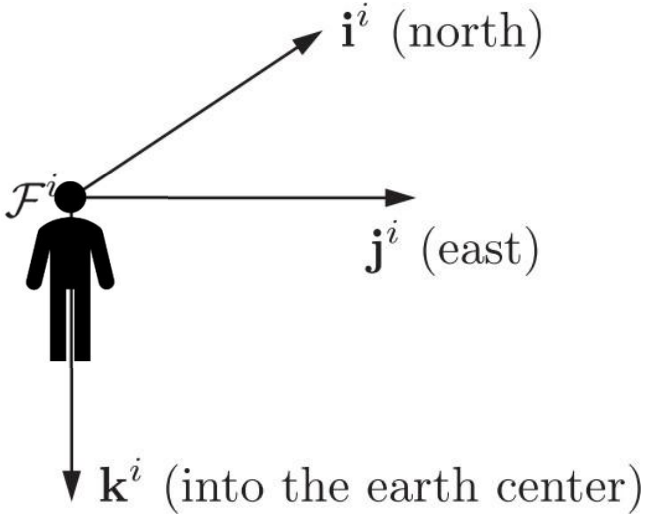
EEEE4138 Aerial Robotics

Tutorial 1 : Reference Frames and Orientation



Reference Frames & Orientation Tutorial



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Question 1: Rotation sequences

How many three-angle axis sequences are available when consecutive rotations use different axes?

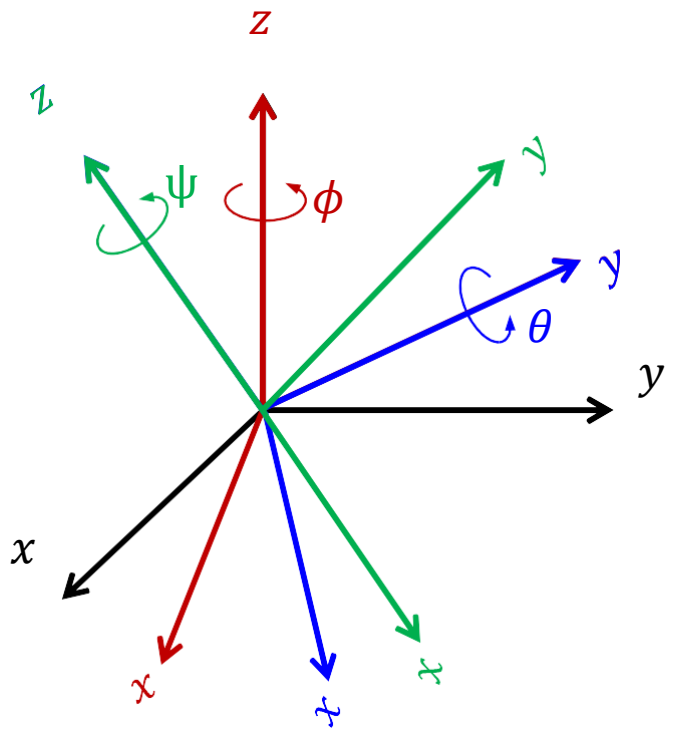
List the proper Euler sequences and the Tait–Bryan sequences.

Explain why consecutive rotations about the same axis are redundant in a three-angle parameterisation.

Question 2: ZYZ at θ = 0

Intrinsic ZYZ sequence:

1. Rotate ϕ about z.
2. Rotate θ about the new y-axis.
3. Rotate ψ about the new z-axis.



Is θ = 0 a valid orientation? Justify your answer.

Can ϕ and ψ then be recovered separately from the matrix?

ZYZ rotation matrix

Intrinsic ZYZ orientation, expressed in the original frame.

$$R = Rz(\varphi)\, Ry(\theta)\, Rz(\psi)$$

$$\begin{bmatrix} c\varphi & -s\varphi & 0 \\ s\varphi & c\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \times \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} c\varphi\, c\theta\, c\psi - s\varphi\, s\psi & -c\varphi\, c\theta\, s\psi - s\varphi\, c\psi & c\varphi\, s\theta \\ s\varphi\, c\theta\, c\psi + c\varphi\, s\psi & -s\varphi\, c\theta\, s\psi + c\varphi\, c\psi & s\varphi\, s\theta \\ -s\theta\, c\psi & s\theta\, s\psi & c\theta \end{bmatrix}$$

cα = cos α, sα = sin α. R maps final-frame components to original-frame components.

Question 3: Recovering ZYZ angles

Use the ZYZ matrix on the preceding reference slide.

Given its entries R_{ij}, derive expressions for ϕ, θ and ψ.
Use the branch $0 < \theta < 180°$.

Use atan2(y, x) to identify the correct quadrant.

State why the expressions fail when $\sin\theta = 0$.

UAV Coordinate Frames Rotations

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Vehicle-to-body coordinate transformation

$$p^b = R^b{}_v \, p^v$$

ϕ = roll, θ = pitch, ψ = yaw. Use a yaw–pitch–roll attitude.

$$R^b{}_v = Rx(-\phi)\, Ry(-\theta)\, Rz(-\psi)$$

$$\begin{bmatrix} c\theta\, c\psi & c\theta\, s\psi & -s\theta \\ s\phi\, s\theta\, c\psi - c\phi\, s\psi & s\phi\, s\theta\, s\psi + c\phi\, c\psi & s\phi\, c\theta \\ c\phi\, s\theta\, c\psi + s\phi\, s\psi & c\phi\, s\theta\, s\psi - s\phi\, c\psi & c\phi\, c\theta \end{bmatrix}$$

$c\alpha = \cos\alpha$, $s\alpha = \sin\alpha$. R_{ij} denotes row i, column j.

This matrix converts vehicle-frame components into body-frame components. It is the inverse of the body-to-vehicle rotation.

Question 4: Recovering roll, pitch and yaw

Use the vehicle-to-body matrix on the preceding slide.

Derive expressions for ϕ, θ and ψ from its entries R_{ij}.

First use the principal pitch range $-90° < \theta < 90°$.
Use atan2(y, x) for roll and yaw.

Then describe the second pitch branch and state when these recovery expressions become singular.

Question 5: Pitch of +90°

Set $\theta = +90°$ in the vehicle-to-body rotation matrix.

Simplify the matrix using angle-difference identities.

What can you determine about roll ϕ and yaw ψ?
Can you recover both angles uniquely?

Give a relation between ϕ and ψ using the matrix entries.

Question 6: Numerical rotation

Use the vehicle-to-body rotation convention.

$$(\phi, \theta, \psi) = (120°, 90°, -60°)$$

Calculate the rotation matrix.

Can the matrix uniquely recover the original three angles? Explain your answer.

Give a different roll–pitch–yaw triple for the same matrix.